

Analysis Note

AI Exposure and Metropolitan Migration: Event Study Design and Results

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1 Data Used in Analysis

The analysis uses the metro-year panel `metro_year_panel.csv` (located in `Data/Processed/`), constructed from IPUMS ACS microdata (2014–2024), FHFA house price indices, and a metro-level AI exposure index. Full details of data sources and construction are described in the companion data note.

The panel covers 242 metropolitan areas observed across up to 11 survey years (2014–2024). Three sample specifications are used:

- **All years** (2014–2024; $N = 2,662$): baseline specification retaining all survey years.
- **Drop 2020** (2014–2019, 2021–2024; $N = 2,420$): excludes the 2020 ACS, which used experimental weighting procedures.
- **Drop 2020 & 2021 (main)** (2014–2019, 2022–2024; $N = 2,178$): additionally excludes 2021 as a COVID-19 recovery year with atypical migration patterns.

The treatment variable is `ai_std`, the standardised metro-level AI exposure index. It is time-invariant by construction and has mean zero and unit standard deviation across the

242 metros (raw range: $[-0.465, 0.232]$, s.d. = 0.125). All regressions are estimated using `feols` from the `fixest` package in R.

2 Econometric Specification

2.1 Treatment Timing

The treatment onset is the public release of ChatGPT in November 2022. Because the ACS records migration in the 12 months prior to the survey date (approximately July of the survey year), the 2022 wave primarily reflects migration behaviour from mid-2021 to mid-2022—before ChatGPT’s release—and is therefore classified as pre-treatment. The post-treatment period covers 2023 and 2024. The base year is 2019.

2.2 Event Study

The primary estimating equation is a continuous-treatment event study (staggered-free, since treatment intensity is pre-determined):

$$Y_{mt} = \alpha_m + \alpha_t + \sum_{k \neq 2019} \beta_k (\widetilde{\text{AI}}_m \times \mathbf{1}[t = k]) + \mathbf{X}'_{mt} \gamma + \varepsilon_{mt} \quad (1)$$

where Y_{mt} is an outcome for metro m in year t ; α_m and α_t are metro and year fixed effects; $\widetilde{\text{AI}}_m$ is the standardised AI exposure score; $\mathbf{X}_{mt}$ is the vector of time-varying controls (share BA+, log population, mean age, female labour force share, unemployment rate); and ε_{mt} is the error term. Standard errors are clustered at the metro level. The base year is 2019.

Pre-treatment coefficients ($k \in \{2014, \dots, 2018\}$ and $k = 2022$ in the main specification) test the parallel-trends assumption; post-treatment coefficients ($k \in \{2023, 2024\}$) capture the causal effect of interest. Event-study plots are saved to `output/` under sub-directories for each specification (`all_years/`, `drop_2020/`, `drop_2020_2021/`).

2.3 Difference-in-Differences Robustness

As a robustness check, the event study is complemented by a simple pre/post difference-in-differences specification:

$$Y_{mt} = \alpha_m + \alpha_t + \delta (\widetilde{\text{AI}}_m \times \text{Post}_t) + \mathbf{X}'_{mt} \gamma + \varepsilon_{mt} \quad (2)$$

where $\text{Post}_t = \mathbf{1}[t \geq 2023]$. The coefficient δ summarises the average post-treatment effect across 2023 and 2024 in a single estimate.

3 Results

3.1 Post-Treatment Event Study Coefficients

Table 1 reports the post-treatment interaction coefficients ($\hat{\beta}_{2023}$ and $\hat{\beta}_{2024}$) for each outcome and specification. Results are from `output/results.txt`.

Table 1: Post-treatment event study coefficients by specification

Outcome	All years		Drop 2020		Drop 2020 & 2021	
	2023	2024	2023	2024	2023	2024
In-migration rate	0.0003	−0.0001	0.0003	−0.0001	0.0002	−0.0001
In-migration rate (BA+)	−0.0006	−0.0010**	−0.0006	−0.0010*	−0.0006	−0.0010
In-migration rate (no BA)	0.0008	0.0009	0.0008	0.0009	0.0008	0.0009
Skilled in-migrant share	−0.0246**	−0.0255**	−0.0244**	−0.0254**	−0.0245**	−0.0255**
Net migration BA+ (rate)	−0.0025**	−0.0017**	−0.0025**	−0.0017**	−0.0025**	−0.0017**
Net migration no BA (rate)	−0.0006	−0.0002	−0.0006	−0.0002	−0.0006	−0.0002
Incumbent share BA	0.0006*	0.0008**	0.0006*	0.0008**	0.0006*	0.0008**
In-mig rate young (25–35)	−0.0003	−0.0000	−0.0003	−0.0000	−0.0003	−0.0000
In-mig rate old (45–55)	0.0004	0.0003	0.0004	0.0003	0.0004	0.0003

Notes: Each cell reports $\hat{\beta}_k$ from Equation (1). Standard errors clustered at the metro level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The treatment variable is the standardised AI exposure index (`ai_std`); coefficients represent the effect of a one-standard-deviation increase in AI exposure.

3.2 Difference-in-Differences Summary

Table 2 reports the pooled DiD coefficient $\hat{\delta}$ from Equation (2) for the three specifications.

Table 2: Difference-in-differences estimates

Outcome	All years	Drop 2020	Drop 2020 & 2021
In-migration rate	0.00049 (0.00056)	0.00047 (0.00057)	0.00023 (0.00056)
In-migration rate (BA+)	−0.00046 (0.00031)	−0.00051 (0.00033)	−0.00068 (0.00032)**
In-migration rate (no BA)	0.00095 (0.00042)**	0.00098 (0.00043)**	0.00090 (0.00043)**
Skilled in-migrant share	−0.01518 (0.00630)**	−0.01520 (0.00645)**	−0.01616 (0.00663)**
Net migration BA+ (rate)	−0.00135 (0.00063)**	−0.00147 (0.00066)**	−0.00162 (0.00065)**
Net migration no BA (rate)	0.00056 (0.00057)	0.00063 (0.00058)	0.00043 (0.00060)
Incumbent share BA	0.00051 (0.00022)**	0.00052 (0.00023)**	0.00061 (0.00023)***
In-mig rate young (25–35)	0.00031 (0.00037)	0.00028 (0.00037)	0.00019 (0.00038)
In-mig rate old (45–55)	0.00014 (0.00022)	0.00016 (0.00022)	0.00011 (0.00022)

Notes: Each cell reports $\hat{\delta}$ (standard error) from Equation (2). $\text{Post}_t = \mathbf{1}[t \geq 2023]$. Standard errors clustered at the metro level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3.3 Summary of Findings

The event study and DiD estimates tell a consistent story across all three specifications. Higher AI-exposed metros see:

- A significant decline in the **skilled in-migrant share** post-treatment ($\hat{\beta}_{2023} \approx -0.025$, $\hat{\beta}_{2024} \approx -0.026$), indicating that the composition of in-migrants shifted away from college-educated workers.
- A significant decline in the **net migration rate among BA+ workers** ($\hat{\beta}_{2023} \approx -0.0025$, $\hat{\beta}_{2024} \approx -0.0017$).

- A modest but significant increase in the **incumbent share of BA+ residents** ($\hat{\beta}_{2024} \approx 0.0008$), consistent with reduced out-mobility among incumbents rather than an increase in in-flows.
- No significant effect on the overall in-migration rate or on non-BA migration.

Pre-treatment coefficients are small and statistically indistinguishable from zero across all outcomes and specifications, supporting the parallel-trends assumption.